C on G (facts, no axioms)

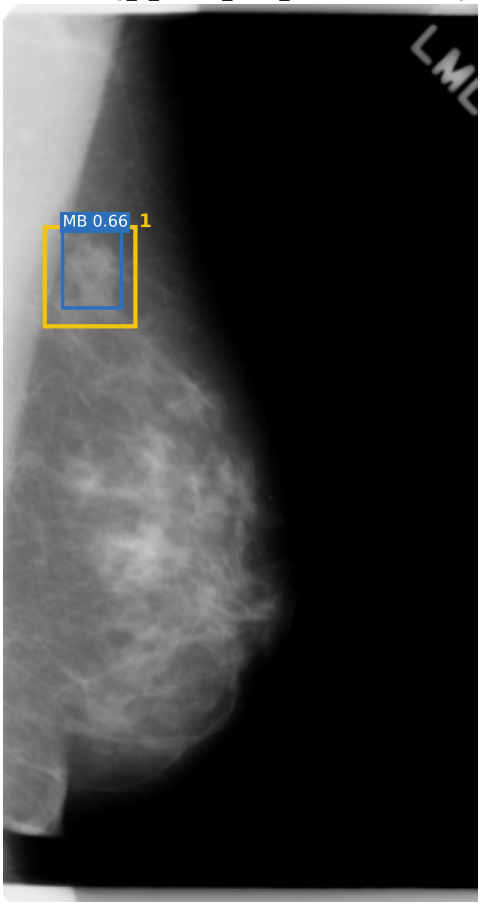
run v5_G_C_geo_e50 best epoch 21 validation mAP@50 0.243

```
Every page: two validation images. Yellow = annotated lesion (number = lesion id).
Solid box = the detection matched to the lesion (highest score with IoU >= 0.5);
dashed = other detections with score >= 0.3. Next to the image, per lesion:
GT = annotation (class = biopsy pathology of the image; descriptors from the CBIS CSV),
PRED = what the model says for the matched box; AXIOMS = rules of the lesion's family whose
antecedent holds, with the truth of the rule on the model's own predictions (ok >= 0.5).

Concept evaluation on ALL validation lesions (matched detections):
   lesions: 438
   detected (IoU >= 0.5): 0.578
   class accuracy: 0.715
   malignancy accuracy: 0.636
   mass_shape accuracy: 0.697
   mass_shape macro-F1: 0.457
   mass_shape majority baseline: 0.572
   mass_margin accuracy (per value): 0.782
   mass_margin macro-F1: 0.000
   mass_margin majority baseline: 0.782
   mass_density accuracy: 0.689
   mass_density macro-F1: 0.626
   mass_density majority baseline: 0.570
   calc_morph accuracy (per value): 0.800
   calc_morph macro-F1: 0.273
   calc_morph majority baseline: 0.776
   calc_distr accuracy: 0.587
   calc_distr macro-F1: 0.148
   calc_distr majority baseline: 0.587
```

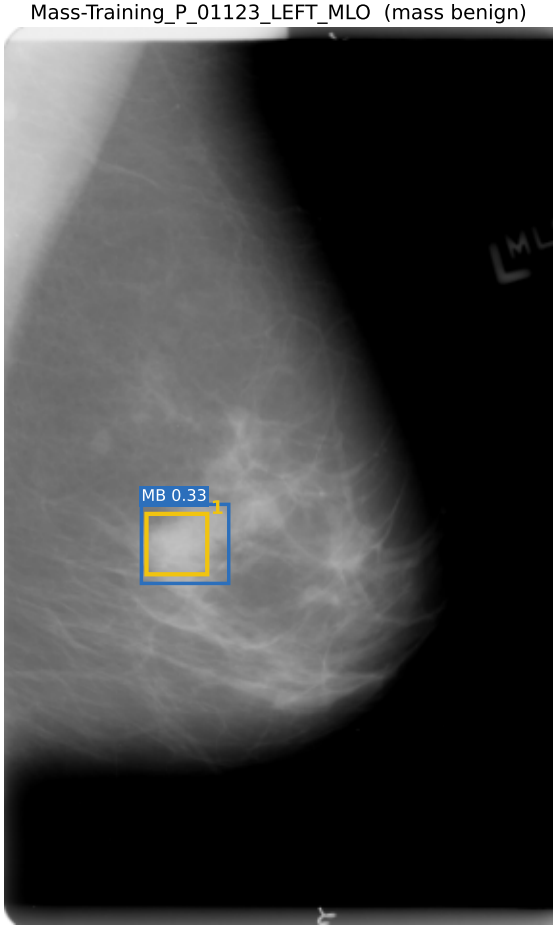
C on G (facts, no axioms)

Mass-Training_P_01052_LEFT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: irregular;  margin: indistinct;  density: low
PRED mass benign 0.66 (malignant 0.24, box score 0.66)
     shape: oval 0.67;  margin: none >= 0.5 (top: circumscribed 0.33);
       density: low 0.79
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.84 ok
     psi5 (mass indistinct -> malignant): 0.75 ok
     psi10 (mass spiculated -> malignant): 0.76 ok
```

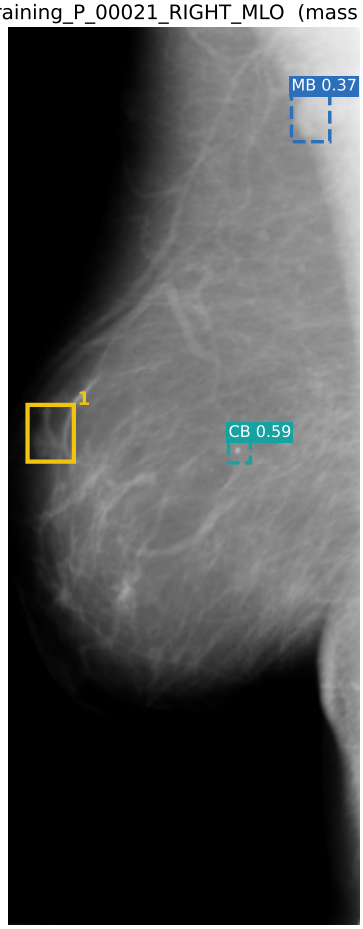
Mass-Training_P_01123_LEFT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed + indistinct;  density: high
PRED mass benign 0.33 (malignant 0.12, box score 0.33)
     shape: oval 0.72;  margin: none >= 0.5 (top: circumscribed 0.38);
       density: low 0.76
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.66 ok
     psi5 (mass indistinct -> malignant): 0.72 ok
     psi10 (mass spiculated -> malignant): 0.68 ok
```

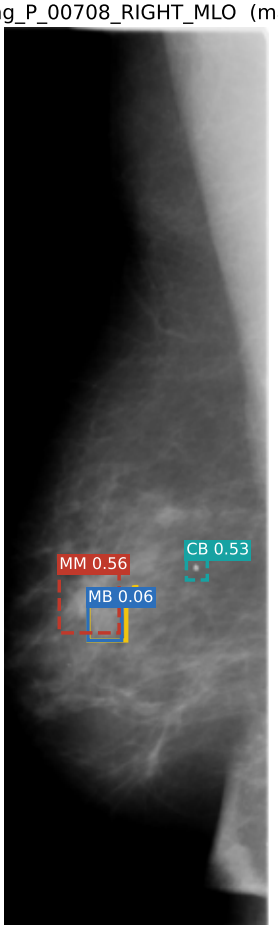
C on G (facts, no axioms)

Mass-Training_P_00021_RIGHT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed;  density: low
PRED not detected (no box with IoU >= 0.5)
```

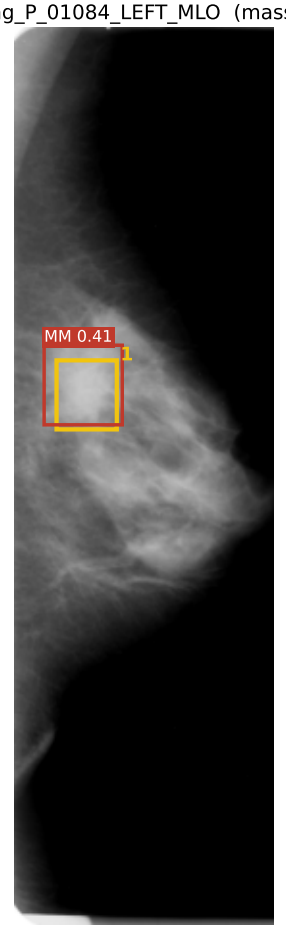
Mass-Training_P_00708_RIGHT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed;  density: low
PRED mass benign 0.06 (malignant 0.01, box score 0.06)
     shape: oval 0.79;  margin: none >= 0.5 (top: circumscribed 0.42);
       density: low 0.92
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.47 VIOLATED
     psi5 (mass indistinct -> malignant): 0.67 ok
```

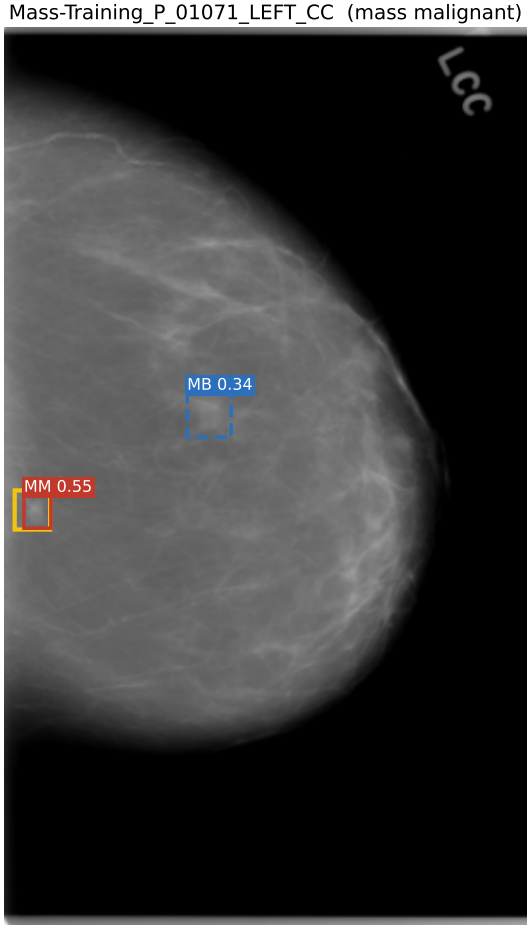
C on G (facts, no axioms)

Mass-Training_P_01084_LEFT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: obscured + spiculated;  density: high
PRED mass malignant 0.41 (malignant 0.45, box score 0.41)
     shape: irregular 0.56;  margin: none >= 0.5 (top: spiculated 0.45);
       density: high 0.59
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.56 ok
     psi5 (mass indistinct -> malignant): 0.79 ok
     psi6 (mass irregular -> malignant): 0.69 ok
     psi9 (mass high density -> malignant): 0.68 ok
     psi10 (mass spiculated -> malignant): 0.76 ok
```

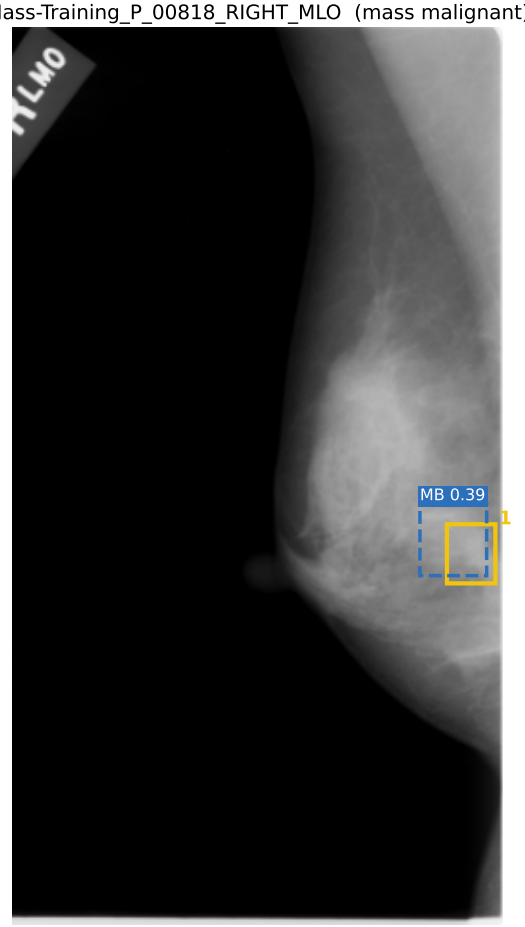
Mass-Training_P_01071_LEFT_CC (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: spiculated;  density: low
PRED mass malignant 0.55 (malignant 0.59, box score 0.55)
     shape: oval 0.50;  margin: none >= 0.5 (top: circumscribed 0.35);
       density: low 0.83
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.61 ok
     psi5 (mass indistinct -> malignant): 0.87 ok
     psi6 (mass irregular -> malignant): 0.82 ok
     psi10 (mass spiculated -> malignant): 0.87 ok
```

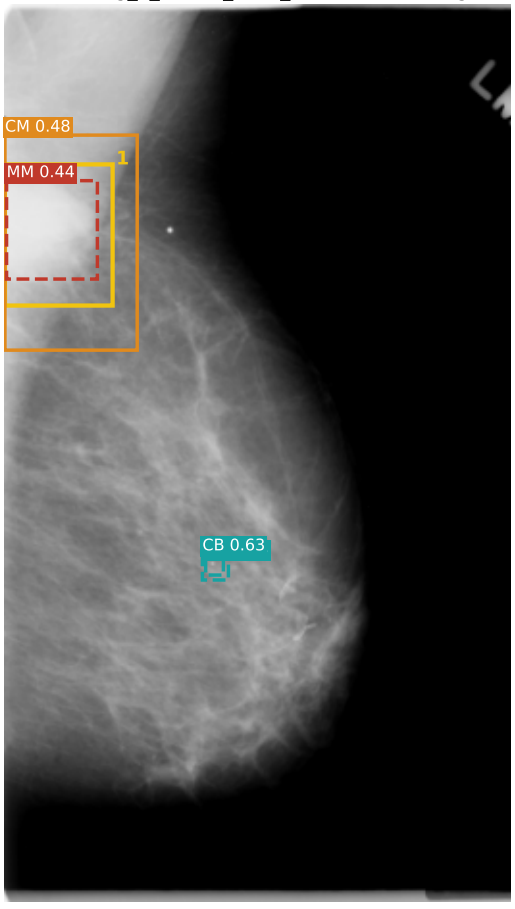
C on G (facts, no axioms)

Mass-Training_P_00818_RIGHT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: spiculated;  density: low
PRED not detected (no box with IoU >= 0.5)
```

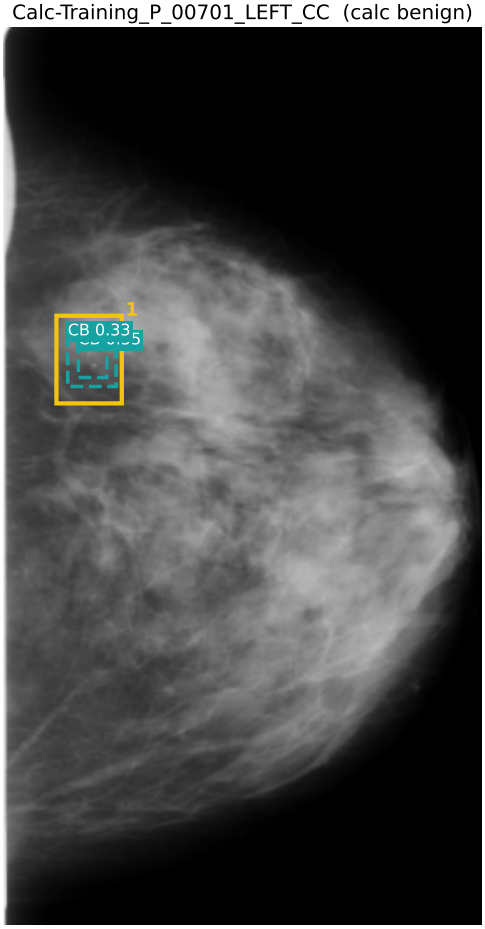
Mass-Training_P_01775_LEFT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: oval;  margin: microlobulated;  density: high
PRED calc malignant 0.48 (malignant 0.49, box score 0.48)
     shape: irregular 0.61;  margin: none >= 0.5 (top: spiculated 0.43);
       density: high 0.67
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.52 ok
     psi5 (mass indistinct -> malignant): 0.83 ok
     psi6 (mass irregular -> malignant): 0.69 ok
     psi9 (mass high density -> malignant): 0.66 ok
     psi10 (mass spiculated -> malignant): 0.78 ok
```

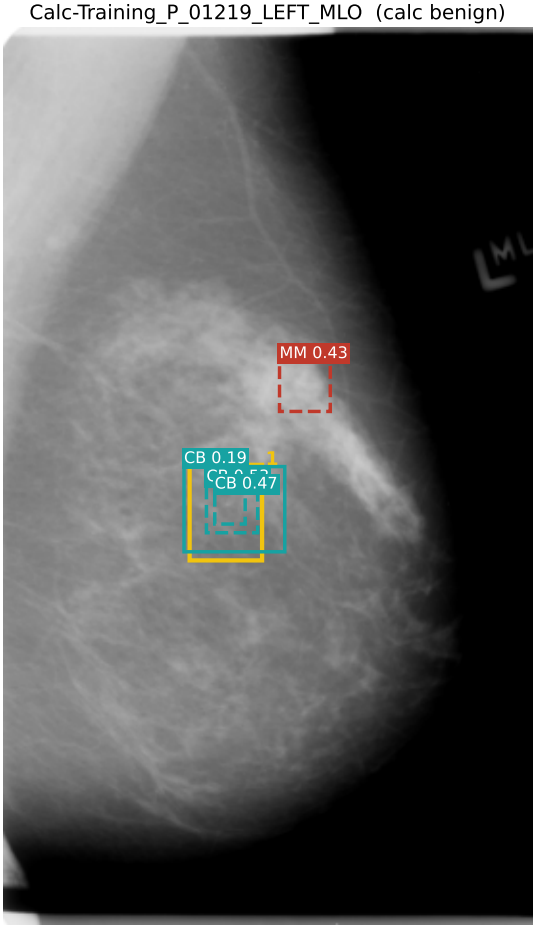
C on G (facts, no axioms)

Calc-Training_P_00701_LEFT_CC (calc benign)




```
Lesion 1
GT   calc benign
     morphology: pleomorphic;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

Calc-Training_P_01219_LEFT_MLO (calc benign)




```
Lesion 1
GT   calc benign
     morphology: pleomorphic;  distribution: clustered
PRED calc benign 0.19 (malignant 0.02, box score 0.19)
     morphology: pleomorphic 0.53;  distribution: clustered 0.96
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.49 VIOLATED
```

C on G (facts, no axioms)

Calc-Training_P_00780_RIGHT_MLO (calc benign)



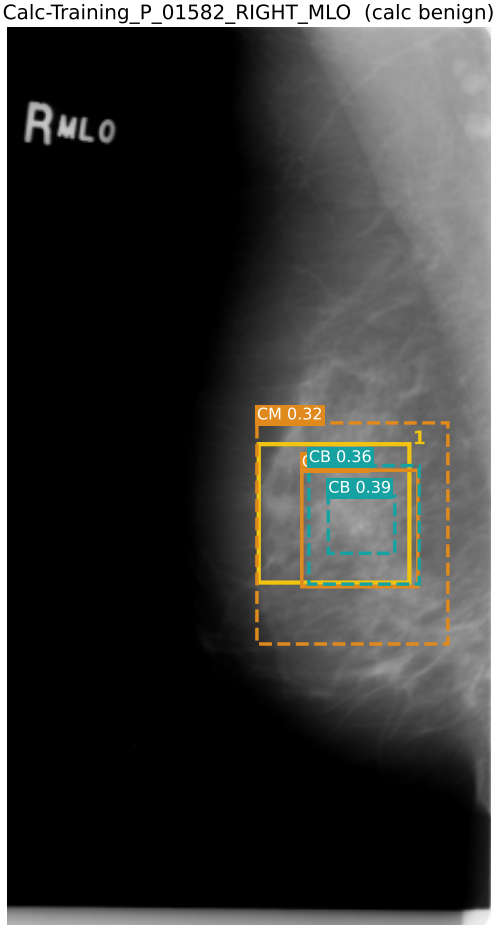

```
Lesion 1
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

Lesion 2
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

Lesion 3
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

(+ 2 more lesions not listed)
```

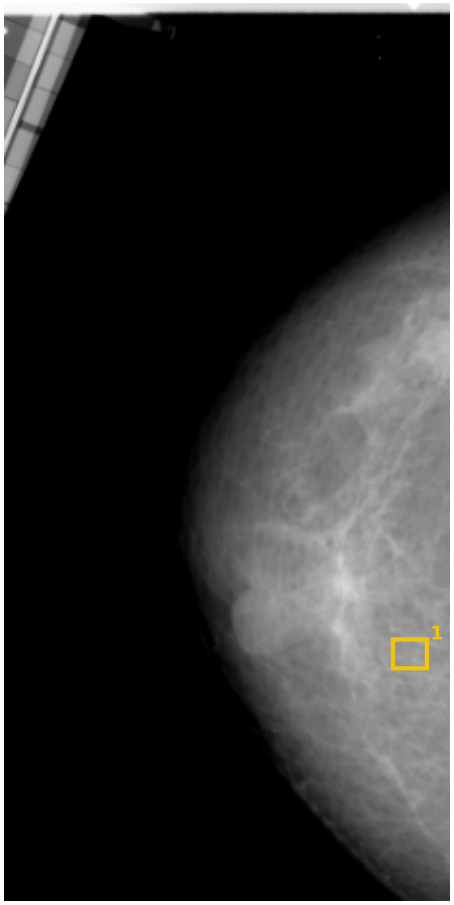
Calc-Training_P_01582_RIGHT_MLO (calc benign)




```
Lesion 1
GT   calc benign
     morphology: amorphous;  distribution: clustered
PRED calc malignant 0.48 (malignant 0.48, box score 0.48)
     morphology: pleomorphic 0.53;  distribution: clustered 0.90
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.74 ok
```

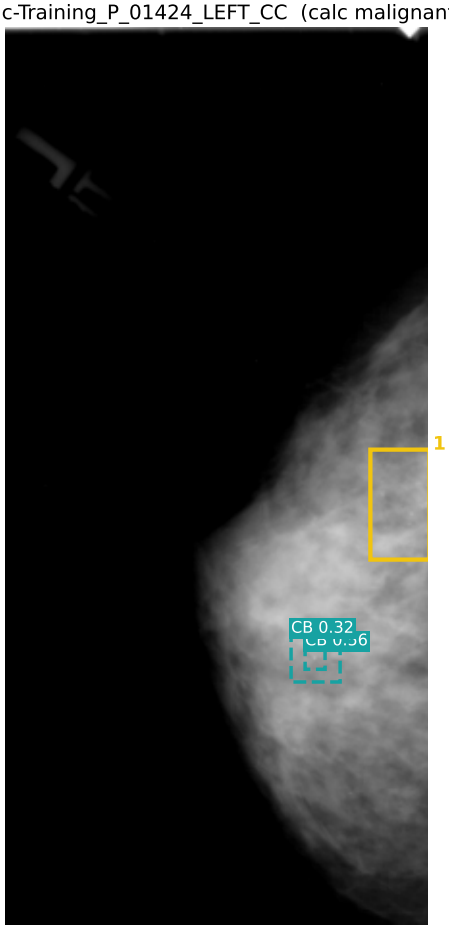
C on G (facts, no axioms)

Calc-Training_P_00735_LEFT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

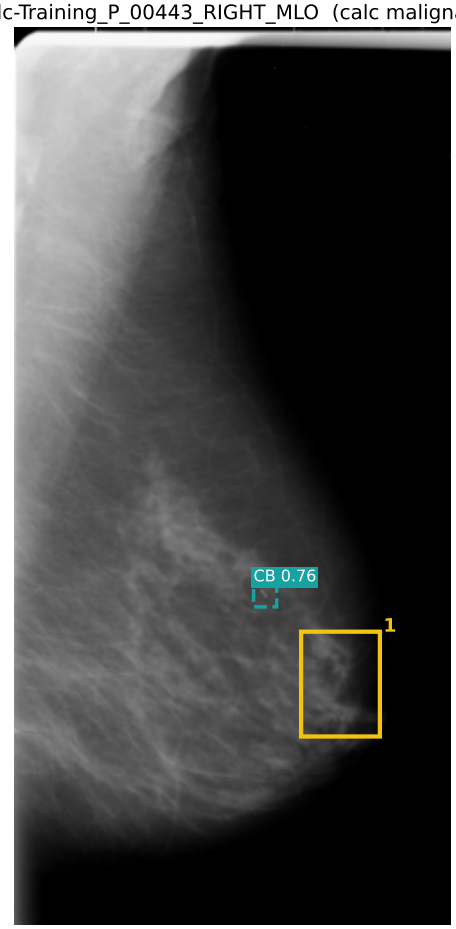
Calc-Training_P_01424_LEFT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

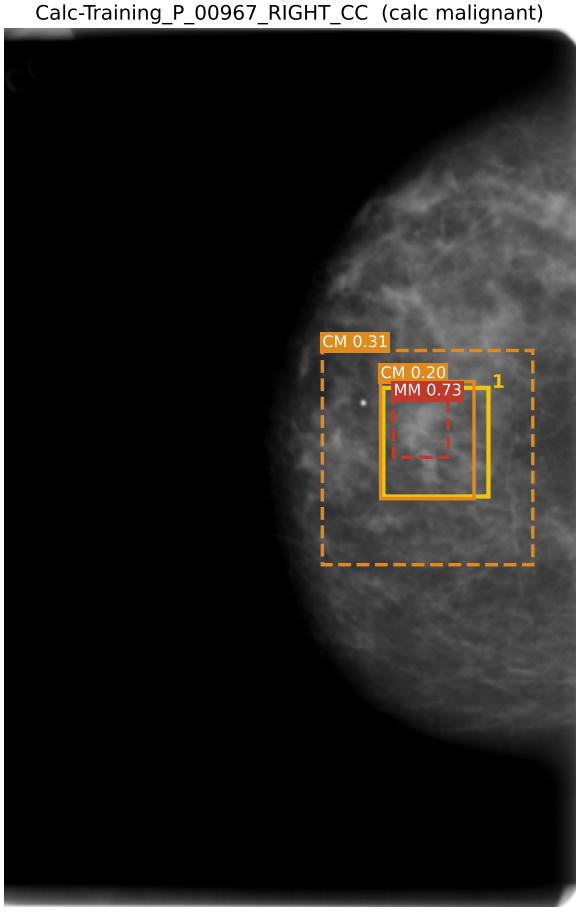
C on G (facts, no axioms)

Calc-Training_P_00443_RIGHT_MLO (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: fine linear branching;  distribution: linear
PRED not detected (no box with IoU >= 0.5)
```

Calc-Training_P_00967_RIGHT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED calc malignant 0.20 (malignant 0.27, box score 0.20)
     morphology: none >= 0.5 (top: pleomorphic 0.49);  distribution:
       clustered 0.88
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.66 ok
```